Panel Feedback

How we responded

IT22090058 **Abinaya R**	✔ **RESOLVED**	
No clear research/tech understanding. Overlap with member 2.	Narrowed scope. Locked predictive APIs (crowd density, weather, sentiment). Distinct component boundary created.	
IT22225092 Wanniarachchi P.M.R	✔ **RESOLVED**	
No clear research/tech understanding. Feasibility needs evaluation.	Mood-based AI architecture defined: BERT emotion classifier + GPT-4 itinerary engine. Feasibility verified via API rate limits and latency tests.	
IT22108654 De Silva D.S.K	✔ **RESOLVED**	
No clear research/tech understanding. Feasibility needs evaluation.	Built multilingual voice SOS pipeline: Whisper STT → danger classifier → Twilio alert. Feasibility confirmed.	
IT22108586 De Silva D.C.M	✔ **RESOLVED**	
No clear research/tech understanding. Lacks research depth and scope for year-long project.	Established verified arrival flow: SLTDA API contract + driver KYC + real-time GPS tracking. Expanded scope to include flight-hotel-driver matcher.	

R26-IT-152 · PP1 RESEARCH PRESENTATION · MAY 2026

AI-Based Tourism Safety & Itinerary Platform for Sri Lanka

SDG 8 · SDG 9 · SDG 11

Supervisors: Prof. Samantha Rajapaksa | Mr. Buddika Harshanath

R26-IT-152 | SLIIT Centre of Excellence for A

I



2.36M Tourists (2025)	**200+** Fatalities Annually	**$1.6B** Revenue Gap (year)	**4** Components Integrated

THE CHALLENGE

The Moment a Tourist Lands at CMB, They're Alone

No safe hotel connection. No verified driver. No crowd warning. No emotional support. No emergency backup.

$148/day Daily spend per tourist	**$300/day** Regional benchmark
1,600+ Guides short (SLTDA)	**72%** Visitors feel overwhelmed

PLATFORM ARCHITECTURE

One Platform. Four Minds. Zero Gaps.



Four specialized AI components — Risk Intelligence, Emotional AI, Emergency SOS, and Safe Arrival — operate as independent services communicating through REST APIs, unified by a shared PostgreSQL database and API bus.

TODAY'S PRESENTATION · 2 MINUTES EACH

Member	Component	Timing
Abinaya R · IT22090058	Tourism Risk & Context Intelligence — Predictive Crowd Risk & Green Sites	2 min
Wanniarachchi P.M.R · IT22225092	Mood-Based AI Travel Companion — Emotional Wellness & Adaptive Itinerary	2 min
De Silva D.S.K · IT22108654	SOS Emergency Detection & Safety Response — Multilingual Voice SOS & Danger Prediction	2 min
De Silva D.C.M · IT22108586	Hotel-Flight Driver Matcher — Verified Arrival & Real-Time Tracking	2 min

SOS SAFETY PLATFORM

Keeping Tourists Safe in Sri Lanka

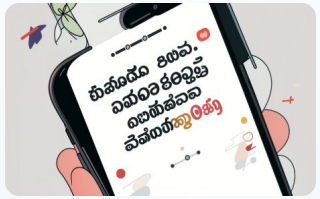
Tourists face harassment, theft, and accidents — with no single emergency number, no early warnings, and language barriers slowing help. Our platform combines a **one-tap SOS** (voice or text, in English, Sinhala, Tamil, and more), an **AI risk map**, and an **automatic WhatsApp alert** with a live map link to a trusted contact — plus the nearest police station and hospital.



TECHNOLOGY & INNOVATION

What Makes It New and Well-Built

A **React front end** and **Flask backend** with 13 endpoints, powered by **three Random Forest models** built with scikit-learn.



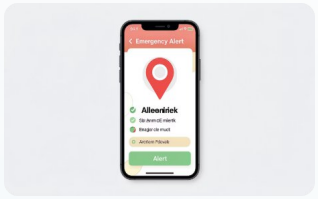
Multilingual Distress Detection

Understands Sinhala, Tamil, and English — not just GPS-based alerts.



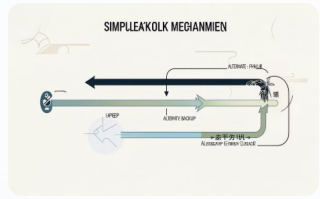
Offline SOS

Alerts are stored and sent automatically when connectivity returns.



Mobile-First UI

One tap opens WhatsApp with a message and live map link.

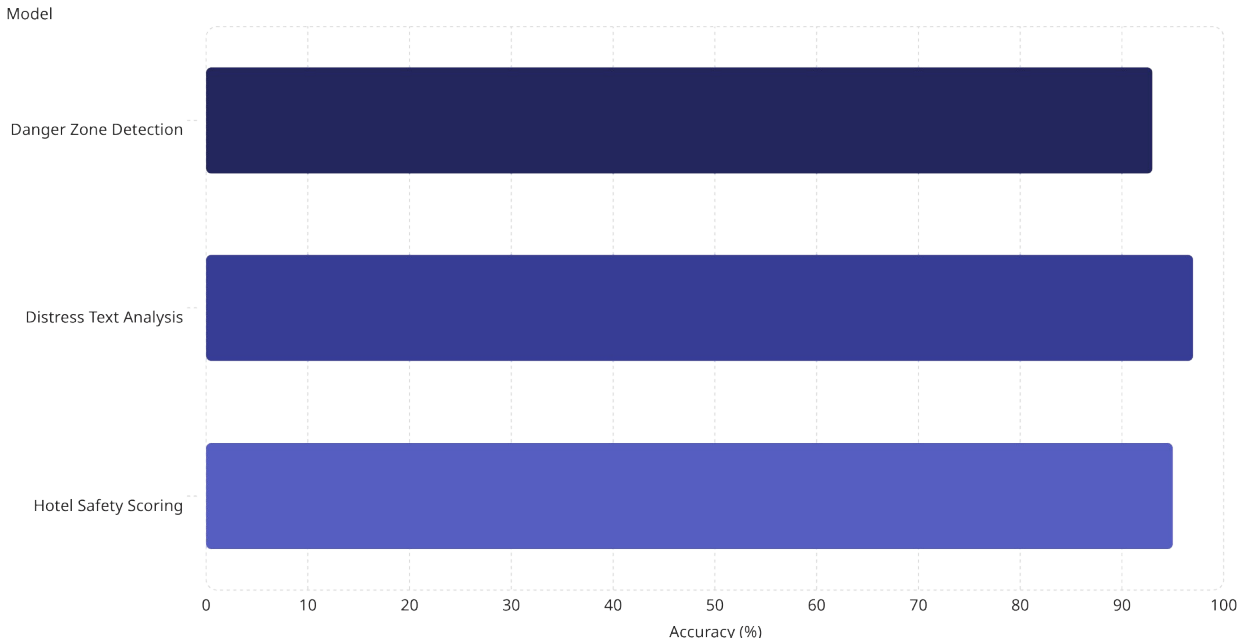


Graceful Degradation

If a model is missing, a rule-based fallback keeps the system running.

PROVEN IMPACT

It Works — and It Matters



Live End-to-End Demo

Voice input → AI analysis → dispatch → WhatsApp alert with map link — all working in real time.

Supporting UN Sustainable Development Goals

SDG 3
Health

SDG 5
Gender Equality

SDG 11
Safer Cities

SDG 16
Stronger Institutions



MoodTour — Emotion-Driven Travel Recommendations (Prototype)

MoodTour reads a traveler's facial expression via webcam, maps that emotion to curated places in Sri Lanka, and returns a ranked day plan that prioritizes safety, mood fit, budget, and traveler type. This slide is your 2-minute opener — clear, compelling, and demo-ready.

Core value

Personalized itineraries that respect how you feel — safer, more delightful travel choices.

Key differentiator

Face emotion directly changes ranking; safety carries 40% weight in the score.

Prototype status

Working local demo: webcam → emotion detection → ranked places → itinerary display.

How It Works — UX Flow & Technical Design



Architecture highlights: modular code (emotion, recommendation, API), Flask REST endpoints, React camera UI, and an end-to-end local demo. Future cloud hosting will swap the local model for an optional secure inference service.

01

1. Capture

Webcam takes a short clip or snapshot in-browser; camera UI validates lighting and face presence.

02

2. Predict

On-device model predicts one of 7 emotions (trained on 28,709 images). Accuracy ≈ 55% vs 14% random.

03

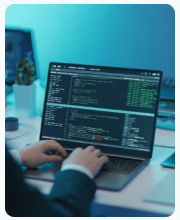
3. Rank

Scoring engine weights Safety (40%), Emotion match (25%), Budget & trip length (20%), Traveler type & prefs (15%).

04

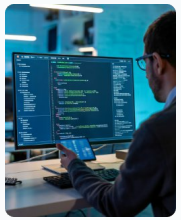
4. Deliver

Frontend shows ranked places with safety scores, emotion fit badges, and a one-day plan. Health check shows API status.



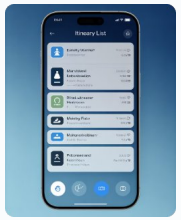
Model & Data

Trained on 28,709 labeled face images for 7 emotions. Single Python script automates training, saving, and exporting the model.



API & Integration

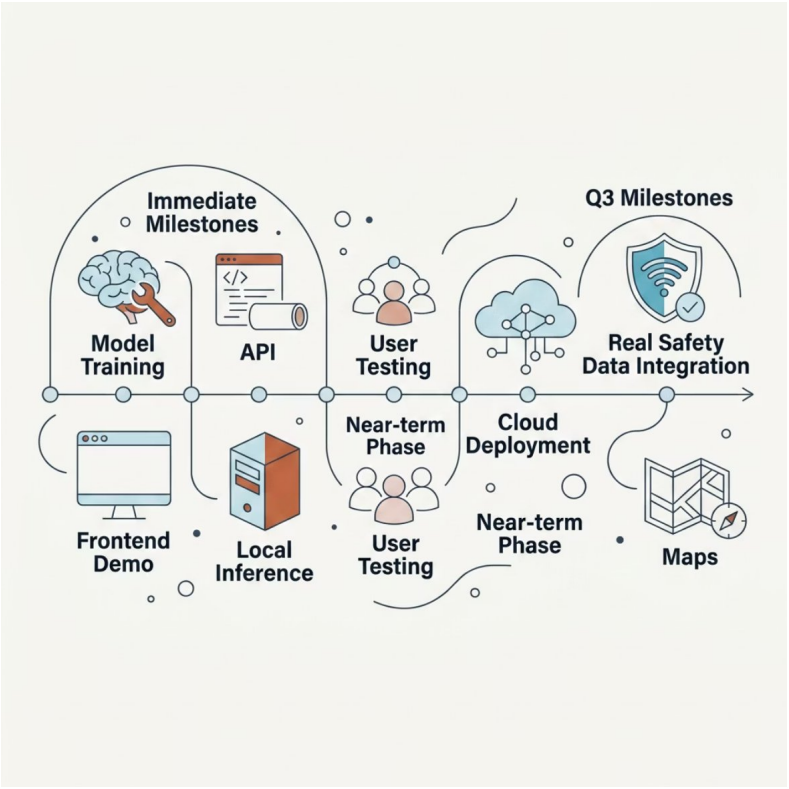
Flask REST API serves predictions and ranking; endpoints include /detect, /rank, and /health with descriptive error messages.



Frontend UX

React UI validates inputs, shows live camera, emotion result, safety indicators, and a generated day plan — ready for live demo.

Validation, Roadmap & Impact



Technical ask & next steps for investors and engineers:

- Fund Phase 2: user testing, safety data licensing, and cloud deployment
- Engineer time: map integration, secure cloud inference, and scale testing
- Pilot target: curated Sri Lanka launch with partner guesthouses and tourism boards

Demo script: Open browser → Click "Detect Emotion" → View mood badge → Click "Generate Itinerary" → Review ranked places and safety-scored day plan.

1

Done

- Model trained & saved (28,709 images)
- Local inference integrated into React demo
- Flask API with documented endpoints
- Scoring system & ranked itineraries

2

Planned (Phase 2)

- Formal user testing and usability studies
- Integrate real crime/safety datasets and dynamic safety scores
- Optional cloud inference and global map integration
- Partnerships with local tourism boards (SDG 17)

3

Impact & SDG Alignment

Supports local economies (SDG 8), fosters innovation (SDG 9), improves safe tourism choices (SDG 11), reduces inequality by supporting varied budgets (SDG 10), and enables partnerships (SDG 17).

R26-IT-152 | AI-BASED TOURISM SAFETY & ITINERARY PLATFORM FOR SRI LANKA

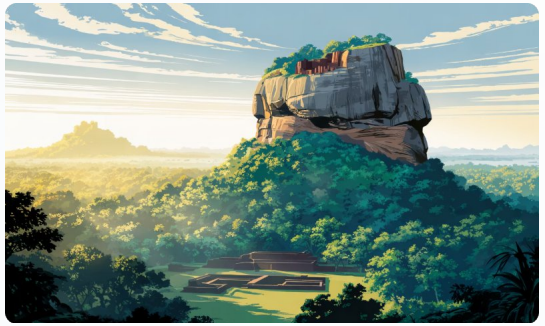
Tourism Risk & Context Intelligence System

Predictive Crowd Risk & Sustainable Site Redirection

Specialization: Information Technology | IT22090058 | Abinaya R | R26-IT-152

2-Minute Delivery Focus

- Predict crowd score (0–1) and risk class (Low/Medium/High) for 30 Sri Lankan tourist sites
- Integrate 4 data sources: AviationStack flights, OpenWeatherMap weather, SLTDA visitor history, group hotel dataset
- Deliver real-time overcrowding alerts via Server-Sent Events (SSE)
- Recommend eco-friendly "green" alternative sites when popular destinations overcrowded
- Expose REST API endpoints (`/risk-score`, `/alert`, `/green-sites`, `/crowd-forecast`) for all 3 teammates to consume



Predicts crowd risk for any Sri Lankan tourist site on any future date with explainable ML (≈91% accuracy) so visitors and authorities can avoid overcrowding and promote sustainable alternatives.

How the Solution Addresses the Sub-Problem

⚠ **PROBLEM:** No existing platform predicts per-site crowd risk for future dates using flight arrivals + weather + holidays in Sri Lanka. Tourists arrive at overcrowded sites blindly, causing stampedes, heat-stroke, scams, and UNESCO heritage damage.

Four-Layer Architecture

01

Data Input Complete

AviationStack CMB flights · OpenWeatherMap · SLTDA 500+ sites · 10K hotel records

02

ML Prediction Complete

Random Forest regressor + classifier · SHAP explainability · 5-fold cross-validation

03

Intelligence Output Complete

Risk badges · time windows · green recommender · SSE alerts

04

API Consumption In Progress

`/risk-score` → Itinerary · `/alert` → SOS · `/green-sites` → Hotel Matcher · `/crowd-forecast` → All modules

Model Performance — All Targets Exceeded

Metric	Target	Achieved	Status
Master dataset	—	21,930 × 27 features	✔
Regression MAE	≤ 0.25	0.0427	✔ PASSED
R^2 Score	≥ 0.70	0.9372	✔ PASSED
Classification accuracy	≥ 80%	≈91%	✔ PASSED
vs. baseline improvement	Significant	78.8%	✔ PASSED

Functional Requirements

ID	Requirement	Status
FR1	Ingest AviationStack, hotel, SLTDA, OpenWeatherMap data	✔ Complete
FR2	Generate crowd score (0–1) classified Low/Medium/High	✔ Complete
FR3	Display recommended visiting time windows	✔ Complete
FR4	Generate SSE real-time alert for High risk	🔄 In Progress
FR5	Dynamic itinerary adjuster with green alternatives	🔄 In Progress
FR6	Green Sites section for eco-friendly alternatives	🔄 In Progress
FR7	REST API endpoints for module integration	✔ Contract agreed
FR8	Anonymous feedback + periodic model retraining	✔ Complete

Design Excellence, Novelty & User Feedback

Novelty Claims

- ✓ First to use CMB flight arrivals as site-level crowd predictor — absent in Bravo et al. (2023), Kim et al. (2025), and 10 reviewed papers
- ✓ First context-aware risk intelligence combining weather + events + holidays + flights for Sri Lanka
- ✓ First green site demand redistribution as working system — Mihalič (2020) theoretical only
- ✓ First dynamic itinerary adjustment linked to real-time risk — Kim et al. (2025) predicted but didn't adjust
- ✓ First post-visit feedback loop in model retraining pipeline

Technical Contribution

- 21,930 records × 27 features × 30 sites master dataset compiled
- Random Forest: MAE 0.0427, R^2 0.9372, ≈91% accuracy — all targets exceeded
- 78.8% improvement over historical average baseline
- SHAP explainability: every prediction auditable and transparent
- Bias audit: no skew against site types, regions, or visitor groups

Validation Status

Completed

- 80/20 train-test split per Bravo et al. (2023) protocol
- 5-fold cross-validation during model training
- API contract agreed with all 3 teammates (March 2026)
- Bias audit completed
- SHAP explainability implemented

Planned

- Field observations at Sigiriya, Galle Fort, Temple of the Tooth — September 2026
- Direct observation vs. predicted crowd levels comparison
- Budget: LKR 15,000 (transport + accommodation)
- Anonymous user testing: 30 diverse travelers — October 2026

SDG Alignment

SDG 11
Sustainable Cities & Communities

SDG 8
Decent Work & Economic Growth

SDG 9
Industry, Innovation & Infrastructure

Abinaya R | IT22090058 | R26-IT-152


THANK YOU!